Relative Strength Index (RSI)

Indicators & Analysis / Momentum & Oscillators | beginner | 5 min read | DeepCharts Help Center | July 15, 2026

The Relative Strength Index (RSI) is one of the most popular technical analysis indicators for evaluating the strength or weakness of an asset over a time period. Developed by J. Welles Wilder, it is a momentum oscillator that moves on a 0–100 scale and is primarily used to identify overbought or oversold conditions, which may indicate a reversal or a consolidation of price.

If you learn one oscillator first, make it this one — most other momentum tools are variations on the same theme.

What it is

RSI compares the size of recent price increases against recent declines over the same window and expresses the result as a single curve between 0% and 100%. When gains have dominated, the curve rises toward the top of the scale; when losses have dominated, it falls toward the bottom.

The DeepCharts version adds an optional smoothed **Average** line on top of the RSI itself, which you can use as a signal line, plus configurable overbought/oversold level lines.

When to use it

- To flag overbought (RSI above the overbought level) and oversold (below the oversold level) conditions that may precede a reversal or pause.
- To read momentum bias: RSI holding in the upper half of its range supports bullish momentum, the lower half bearish.
- To spot divergences: price making a new extreme while RSI does not is an early warning that momentum is fading (see Divergence Detector).
- In ranges, as a mean-reversion timing tool — the environment where the classic 70/30 signals work best.

Quick start

1. Open a chart and click the bar-chart icon in the top-left corner to open the **Indicators** panel.
2. Click **Indicators** to open the full **Indicator List**.
3. Search for "Relative Strength Index" and click **+** to add it — it plots in its own panel below the price chart.
4. Click the gear icon next to the indicator to open its settings.

The defaults are the textbook configuration: **Length** 14 with the **Overbought Level** at 70 and the **Oversold Level** at 30. Leave them as they are while you learn the indicator; they are also what most other traders are watching, which matters for levels that act partly by consensus.

[SCREENSHOT PLACEHOLDER] A price chart with the RSI indicator in a lower panel — the RSI curve oscillating between the 70 and 30 level lines, with a recent excursion above 70 highlighted `rsi-on-chart.png`

How to read it

- **Above the overbought level (default 70)**: recent gains have strongly outweighed losses. This flags stretched momentum — it is a caution sign, not an automatic sell.
- **Below the oversold level (default 30)**: the mirror condition on the downside.
- **The 50 midline**: RSI spending its time above 50 indicates bullish momentum regime; below 50, bearish. Many trend traders use the midline rather than 70/30.

- **Divergence**: price prints a higher high while RSI prints a lower high (bearish divergence), or price a lower low while RSI a higher low (bullish divergence). Divergences warn that the move's momentum is thinning.
- **With the Average line enabled**: RSI crossing above its average is a momentum-turning-up cue, crossing below the opposite — similar to a signal line on MACD.

Warning: In a strong trend, RSI can stay overbought or oversold for a long time while price keeps running. Fading every 70/30 touch in a trending market is the most expensive way to use this indicator.

Settings reference

Parameters — General

Setting	What it does
Input data	Which price data feeds the calculation. See Different Types of Input Data for Indicators.
Length	Number of periods used in the RSI calculation. Default: 14. Shorter lengths make the curve faster and noisier; longer lengths smooth it.

Parameters — Average

Setting	What it does
Enable	Toggles the additional average (signal) line calculated on the RSI.
Average Type	The averaging methodology used for the signal line.
Length	Number of periods for the average. Default: 20.

Parameters — Level settings

Setting	What it does
Overbought Level	The upper threshold line. Default: 70. Raise it (e.g. toward 80) to demand more extreme readings in strongly trending markets.
Oversold Level	The lower threshold line. Default: 30.
Overbought Color / **Oversold Color**	Colors of the two threshold lines.
Level Width	Width setting for the level lines. Default: 10.

Subgraph

Setting	What it does
RSI line color / average line color / secondary color	Colors for the RSI curve, its average line, and the secondary color option.
Display style / line style / line width	How the curves are drawn and how thick.
Secondary axis display	Option to display the indicator on a secondary axis.

[SCREENSHOT PLACEHOLDER] The RSI settings dialog showing the General group (Input data, Length 14), the Average group (Enable, Average Type, Length 20) and the Level settings group (Overbought 70, Oversold 30)

`rsi-settings.png`

Tips and common mistakes

- **Context first.** The same RSI 75 reading is a fade candidate in a range and a strength confirmation in a fresh breakout. Decide what regime you are in — a trend filter like ADX helps — before applying overbought/oversold logic.

• **Divergence needs confirmation.** A divergence is a warning, not an entry; wait for price structure (a break of a swing level) to confirm before acting on it.

• **Don't shorten Length to "see signals earlier".** A 5-period RSI produces constant extreme readings that mean very little; if you need faster signals, shorten your chart timeframe instead and keep the RSI window meaningful.

• **Use the Average line to cut noise**: requiring RSI to be above/below its own average filters a surprising number of false midline crosses.

Related articles

• Stochastic Oscillator

• MACD

• Williams %R

• Divergence Detector

• Different Types of Input Data for Indicators

• How to Change the Layout of Indicators